

G. Lejeune Dirichlet's Werke. II.

XVII.

On a Theorem Concerning Series.

By M. G. Lejeune Dirichlet

English translation

On a Theorem Concerning Series.

The theorem which I am about to establish in a new way is the same one which has served me as a lemma in several preceding memoirs. The proof which I gave of it in the first of these memoirs assumes a certain condition which, although it is satisfied in most of the applications one can make of the lemma, is not indispensable, as I have already remarked elsewhere and as will also be seen in what follows.

I begin with a very simple special case. Let a and b denote positive constants and let ρ likewise be a positive variable. Then

$$\lim_{\rho \rightarrow 0} \rho \sum_{n=0}^{n=\infty} \frac{1}{(b+na)^{1+\rho}} = \frac{1}{a},$$

where the limit refers to the indefinite decrease of ρ . To prove this special case, consider the integral

$$\int_b^{\infty} \frac{dx}{x^{1+\rho}} = \frac{1}{\rho b^{\rho}}$$

as the area under a curve whose abscissa is x . Since this curve approaches more and more closely the x -axis, if one draws parallels to this axis through the endpoints of the ordinates corresponding to $x = b, b+a$, and so on, one obtains exterior and interior rectangles for the area. It follows at once that the sum whose limit is in question lies between the two quantities

$$\frac{1}{ab^{\rho}} \quad \text{and} \quad \frac{1}{ab^{\rho}} + \frac{\rho}{b^{1+\rho}},$$

whose common limit is $1/a$.

Now to pass to the general theorem, let

$$k_1, \quad k_2, \quad k_3, \quad \dots, \quad k_n, \quad \dots$$

be positive constants arranged in increasing order of magnitude, so that $k_n \leq k_{n+1}$. Suppose that these constants are such that, if t denotes a positive continuous variable and T the number of terms of our sequence which do not exceed t , the ratio

$$\frac{T}{t}$$

converges to the finite limit α as t increases indefinitely. Under this assumption, I say that the same quantity α is also the limit of the sum

$$\rho \sum_{n=1}^{n=\infty} \frac{1}{k_n^{1+\rho}} \tag{1}$$

for an infinitely small value of ρ .

By the assumed condition, if δ denotes a number as small as desired, one can choose another number τ large enough so that, whenever t exceeds τ , one always has

$$\alpha - \delta < \frac{T}{t} < \alpha + \delta.$$

This done, let N be the value of T corresponding to $t = \tau$, and consider first only those terms of our series whose index $n > N$. Such a term k_n can present two cases, according as one has

$$k_n < k_{n+1} \quad \text{or} \quad k_n = k_{n+1}.$$

In the first case, the value of T corresponding to $t = k_n$ is n , and one has

$$\alpha - \delta < \frac{n}{k_n} < \alpha + \delta.$$

In the second case, let $k_{m'}$ be the last among the terms following k_n which is still equal to k_n , and, among the preceding terms, let k_m be the first whose value is smaller than that of k_n . If now we let t increase starting from $t = k_m$, T remains constantly equal to m until t reaches the value $k_{m'}$, and our ratio T/t thereby approaches $m/k_{m'}$ indefinitely, while always remaining greater than $\alpha - \delta$. Hence

$$\frac{m}{k_{m'}} > \alpha - \delta.$$

But by the first case we also have

$$\frac{m'}{k_{m'}} < \alpha + \delta,$$

and moreover

$$m < n < m', \quad k_n = k_{m'}.$$

We therefore conclude, as before, that

$$\alpha - \delta < \frac{n}{k_n} < \alpha + \delta.$$

Putting this double inequality in the form

$$(\alpha - \delta)^{1+\rho} \frac{\rho}{n^{1+\rho}} < \frac{\rho}{k_n^{1+\rho}} < (\alpha + \delta)^{1+\rho} \frac{\rho}{n^{1+\rho}},$$

and summing from $n = N + 1$ to $n = \infty$, we obtain

$$(\alpha - \delta)^{1+\rho} \rho \sum_{n=N+1}^{n=\infty} \frac{1}{n^{1+\rho}} < \rho \sum_{n=N+1}^{n=\infty} \frac{1}{k_n^{1+\rho}} < (\alpha + \delta)^{1+\rho} \rho \sum_{n=N+1}^{n=\infty} \frac{1}{n^{1+\rho}}.$$

But the expression $\rho \sum 1/n^{1+\rho}$ falls under the special case examined above when one puts

$$b = N + 1, \quad a = 1,$$

and consequently has unity as its limit. Thus the part of our sum (1) which extends from $n = N + 1$ to $n = \infty$ will eventually remain between $\alpha - \varepsilon$ and $\alpha + \varepsilon$, where ε is only slightly larger than δ and therefore, like δ , arbitrarily small. At the same time the first part, which has only the finite number N of terms, plainly converges to zero. It follows that the limit of expression (1) is indeed α , Q.E.D.